UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	FEBRUARY 2025
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts : PART A (10 Questions)
   PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of :
   i) the Question Paper
   ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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PART A

QUESTION 1

The evolution of computers and technological advancements have led to powerful variations that can process data faster, perform complex calculations, support artificial intelligence, enable global communication, manage vast databases, simulate real-world scenarios, and offer automation, all while being smaller, more energy-efficient, and widely accessible.

a) Name the primary technology in the zero, first, and second computer generations.

(3 marks)

b) List **TWO (2)** fundamental components of integrated circuits.

(2 marks)

QUESTION 2

The availability of personal computers (PCs) impacted daily life, business, and society. Computing became affordable and accessible to individuals and small businesses, allowing modern communication and information sharing. It boosts work productivity by automating tasks and better data processing.

a) In which computer generation that the PC was first introduced?

(1 mark)

b) State the name of the first commercially available computer.

(1 mark)

c) What kind of interface was available when the PC was first introduced?

(1 mark)

d) Discuss the key factor that makes the personal computer portable.

(2 marks)

QUESTION 3

a) Calculate the byte content shown in Figure 1.

(2 marks)

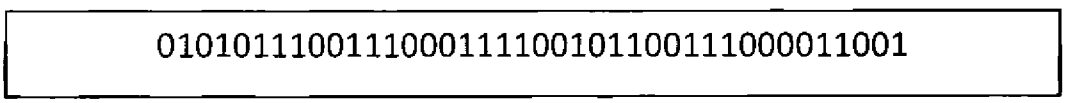


Figure 1

b) Calculate $2D.5_h + ED.2_h$ in binary. Show your complete workflow.

(3 marks)

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QUESTION 4

a) Calculate 40_8 - 40_h using sign and magnitude. Show your complete workflow.

(2 marks)

b) Calculate $39_h - 23_8$ using 1's and 2's complement method. Show your complete workflow.

(3 marks)

QUESTION 5

a) Convert -81.21875 to a floating point number in IEEE single precision format in hexadecimal value form. Show your complete workflow.

(5 marks)

b) Convert 43A37C00h in IEEE single precision floating point number to decimal value form. Show your workflow.

(5 marks)

QUESTION 6

a) Identify the correct concept of computer organization or architecture in the following scenario.

(4 marks)

Scenario	Concept
Deals with the logical aspects of the system, including instruction sets, data formats, and addressing modes.	
Defines the functionality, capabilities, and programming model.	
Concerned with how the computer will execute the tasks and implement the functions.	
Involves the hardware layout, circuits, control signals, memory access, and timing.	

b) Describe **THREE (3)** CPU registers involved in data processing, such as arithmetic operations.

(6 marks)

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QUESTION 7

a) What is a cache?

(2 marks)

b) Elaborate on what will happen if there is no cache in the computer system.

(2 marks)

c) Comparing L1 and L2 cache, discuss which cache would have faster speed with relevant justification.

(2 marks)

d) Discuss **TWO (2)** policies to write data to the cache to ensure consistency between the cache and the main memory (RAM).

(4 marks)

QUESTION 8

a) Compare **TWO (2)** system bus timing approach with appropriate examples.

(6 marks)

b) Discuss **ONE (1)** reason why an I/O module is needed.

(2 marks)

c) List **TWO (2)** interfaces to external devices

(2 marks)

QUESTION 9

a) Differentiate the involvement of CPU in the interrupt-driven I/O technique and Direct Manipulation Access

(4 marks)

b) Define interrupt.

(2 marks)

c) Discuss what happens during a context switch.

(2 marks)

d) Elaborate on the benefit of a context switch.

(2 marks)

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QUESTION 10

a) Draw Single Instruction, Single data (SISD) architecture with the following information:
- CU is the control unit
- PU is the processing unit
- MU is the memory unit
- IS is the instruction stream
- DS is the data stream

(2 marks)

b) Describe the parallel architecture.

(2 marks)

c) Explain **ONE (1)** benefit of parallel architecture.

(2 marks)

d) Differentiate between loosely-coupled and strongly-coupled Multiple Instruction, Multiple Data stream (MIMD) architecture with example.

(4 marks)

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PART B

QUESTION 1

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

One-address	Two-address	Three-address
LOAD p	MOV (p,q)	ADD (p,q,r)
STOR p	ADD (p,q)	SUB (p,q,r)
ADD p	SUB (p,q)	MUL (p,q,r)
SUB p	MUL (p,q)	DIV (p,q,r)
MUL p	DIV (p,q)	
DIV p		

(20 marks)

END OF QUESTION PAPER

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